LAB 1: Introduction to Numerical Methods

Objective: To understand basics of numerical methods for engineers.

Theory:

Numerical methods are techniques by which mathematical problems are formulated so that they can be solved with arithmetic and logical operations. The implementation of a numerical method with an appropriate convergence check in a programming language is called a numerical algorithm.

In the pre–computer era, the time and work of implementing such calculations seriously limited their practical use. However, with the advent of fast, inexpensive digital computers, the role of numerical methods in engineering and scientific problem solving has exploded.

Example – Suppose a bungee-jumper is falling freely as shown in the figure-

The governing equation based on newton's laws are given as below,

$$F = ma = m\frac{dv}{dt}$$

And *net* $F = mg - F_{drag}$

$$m\frac{dv}{dt} = mg - F_{drag}$$

$$\frac{dv}{dt} = g - \frac{F_{drag}}{m} = g - \frac{C_d V^2}{m}$$

Where, v = velocity of falling man, g is acceleration due to gravity = 9.81, cd = drag coefficient = 0.25, m = mass of the bungee jumper = 68.1 kg, t = time.

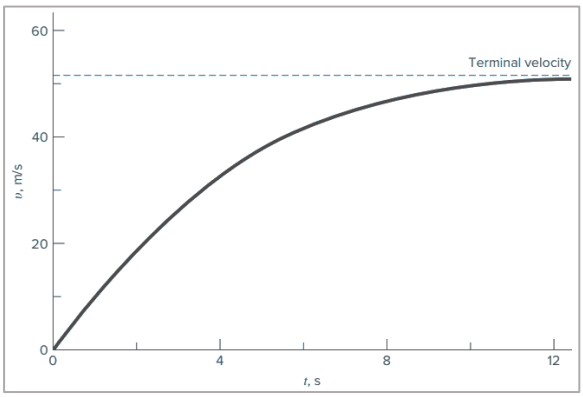
$$\frac{dv}{dt} = g - \frac{c_d}{m}v^2$$

if the jumper is initially at rest (υ = 0 at t = 0), calculus can be used to solve equation with varying t values.

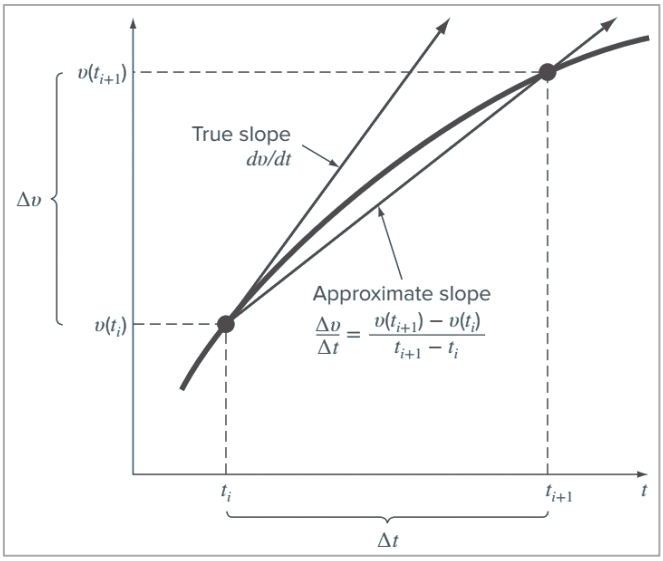
$$v(t) = \sqrt{\frac{gm}{c_d}}\tanh\left(\sqrt{\frac{gc_d}{m}}t\right)$$

t, s	v, m/s
0	0
2	18.7292
4	33.1118
6	42.0762
8	46.9575
10	49.4214
12	50.6175
∞	51.6938

Plot the Solution →



Alternatively, we can solve above problem numerically. The use of a finite difference to approximate the first derivative of υ with respect to t.



$$\frac{dv}{dt} \cong \frac{\Delta v}{\Delta t} = \frac{v(t_{i+1}) - v(t_i)}{t_{i+1} - t_i}$$

$$v(t_{i+1}) = v(t_i) + \left[g - \frac{c_d}{m} v(t_i)^2\right](t_{i+1} - t_i)$$

We can now see that the differential equation has been transformed into an equation that can be used to determine the velocity algebraically at t_{i+1} using the slope and previous values of v and t. If you are given an initial value for velocity at some time t_i, you can easily compute velocity at a later time t_{i+1}. This new value of velocity at t_{i+1} can in turn be employed to extend the computation to velocity at t_{i+2} and so on. Thus at any time along the way,

New value = old value + slope × step size

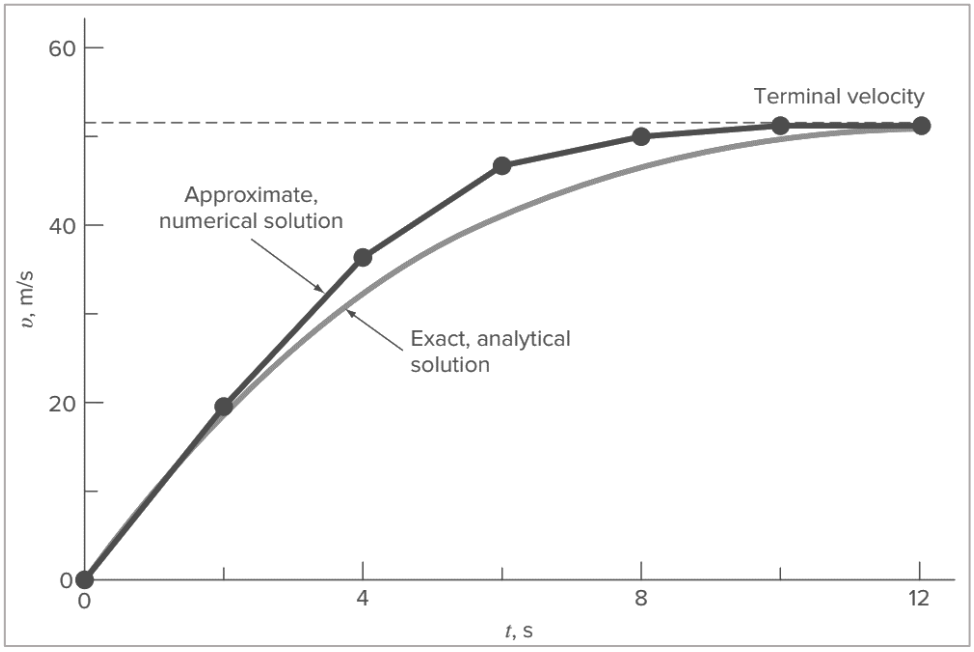
This approach is formally called **Euler's method**. Employ a step size of 2 s for the calculation, and initial condition as (υ = 0 at t = 0).

Solution. At the start of the computation ($t_0 = 0$), the velocity of the jumper is zero. Using this information and the parameter values from Example 1.1, Eq. (1.12) can be used to compute velocity at $t_1 = 2$ s:

$$v = 0 + \left[9.81 - \frac{0.25}{68.1}(0)^2\right] \times 2 = 19.62 \text{ m/s}$$

For the next interval (from $t = 2$ to 4 s), the computation is repeated, with the result

$$v = 19.62 + \left[9.81 - \frac{0.25}{68.1}(19.62)^2\right] \times 2 = 36.4137 \text{ m/s}$$



Few other real physical problems,

Assignment -

Any numerical as suggested by Faculty.

Experiment: Bungee Jumper Velocity Analysis using Euler's Method

Aim:

To compute the velocity of a bungee jumper using Euler's numerical method and validate the numerical solution by comparison with the analytical solution.

Calculations:

Q1. Free falling bungee jumper

* Analytical solution:

Governing equation :- $\frac{dv}{dt} = g - \frac{C_d}{m} v^2$

Using calculus to obtain:

$$v = \sqrt{\frac{gm}{C_d}} \cdot \tanh\left(\sqrt{\frac{gC_d}{m}}\, t\right)$$

* Numerical solution :-

Governing eq : $\frac{dv}{dt} = g - \frac{C_d}{m} v^2$

Approximating $\frac{dv}{dt}$ as $\frac{\Delta v}{\Delta t}$

And using Euler's method to obtain

$$V_{n+1} = V_n + \left[g - \frac{C_d}{m} V_n^2\right](\Delta t)$$

Where, $\frac{\Delta v}{\Delta t} = \frac{V_{n+1} - V_n}{t_{n+1} - t_n}$

Given Parameters:

g	9.81
Mass	80
Cd	0.25

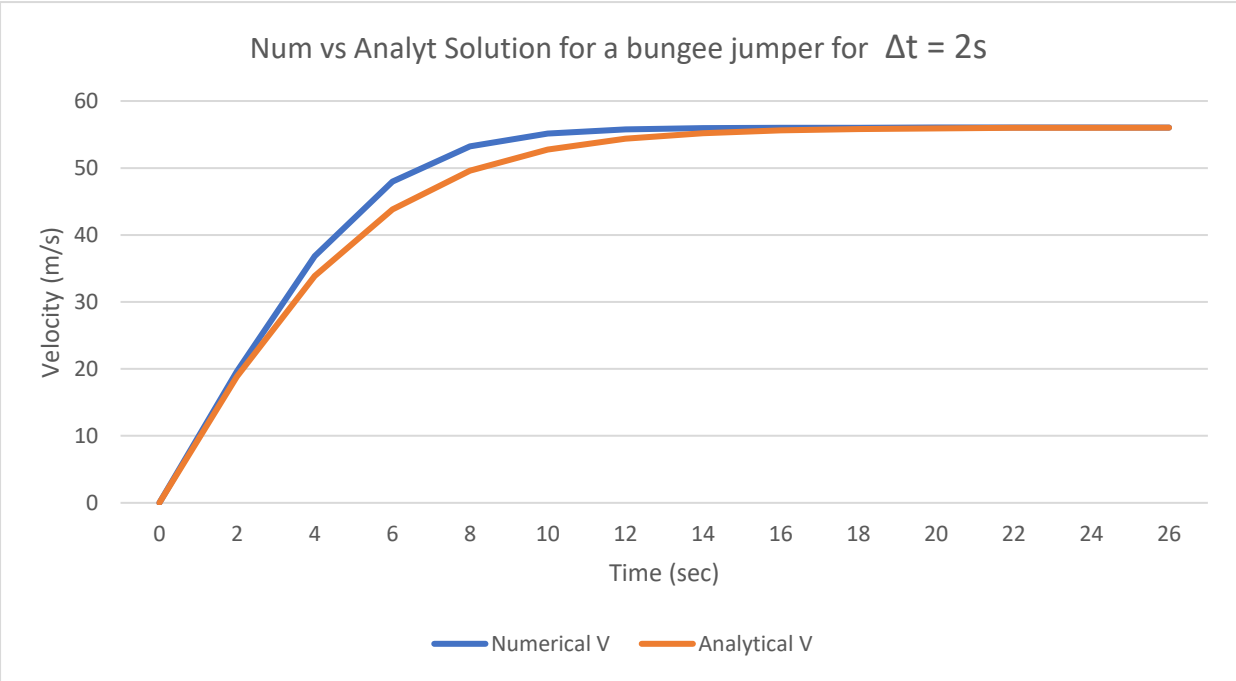
Calculation Table:

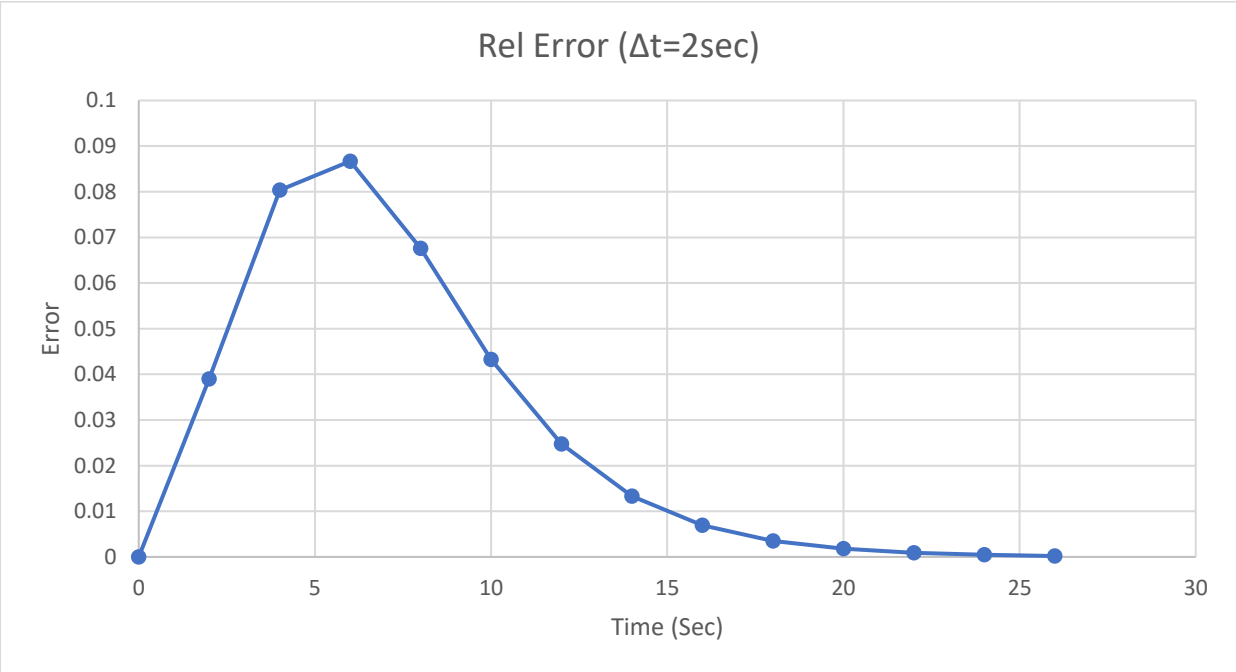
For Timestep or Delta t = 2secs

Time	Numerical V	Analytical V	Rel Error (Δt=2sec)
0	0	0	0
2	19.62	18.85550901	0.038964882
4	36.8340975	33.87455406	0.080347929
6	47.97440538	43.81514911	0.086697401
8	53.20975806	49.61362445	0.0675841
10	55.13426835	52.74960702	0.04325189
12	55.75559619	54.37651922	0.024734324
14	55.94630552	55.20234158	0.013297821
16	56.00387364	55.61689205	0.006909908
18	56.021162	55.82382803	0.00352249
20	56.0263458	55.9268379	0.001776091
22	56.0278994	55.97804332	0.000889844
24	56.02836496	56.00347951	0.000444158
26	56.02850446	56.01611052	0.000221208
28	56.02854626	56.02238172	0.000110025
30	56.02855878	56.02549505	5.46816E-05

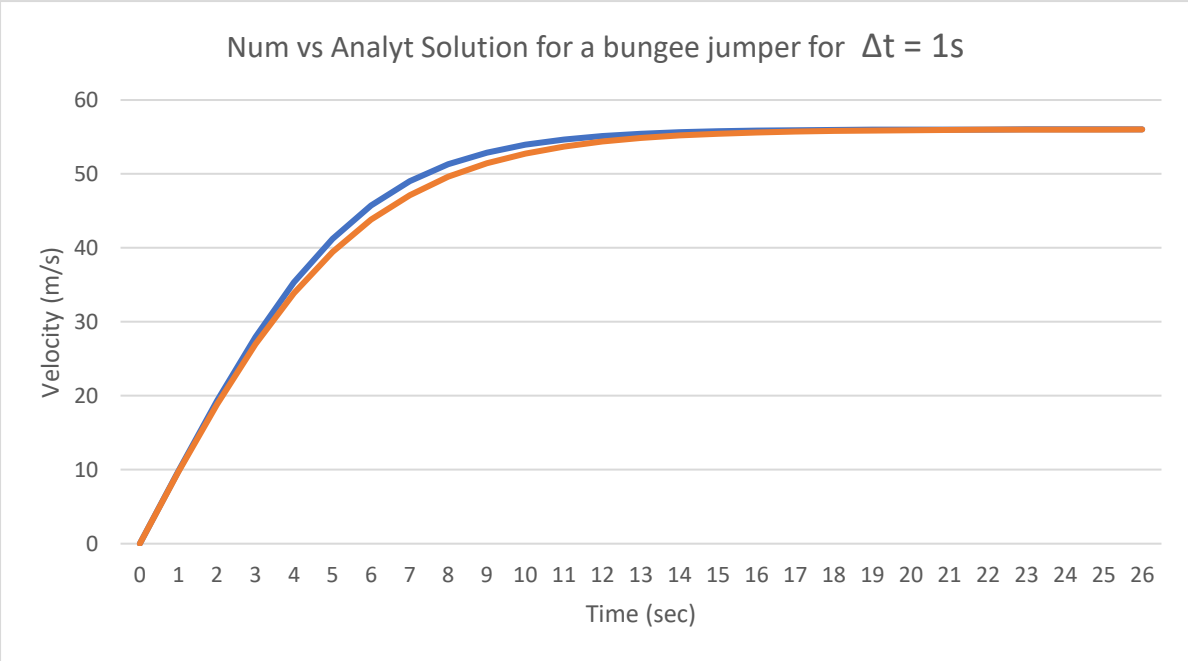
Graphical Analysis:

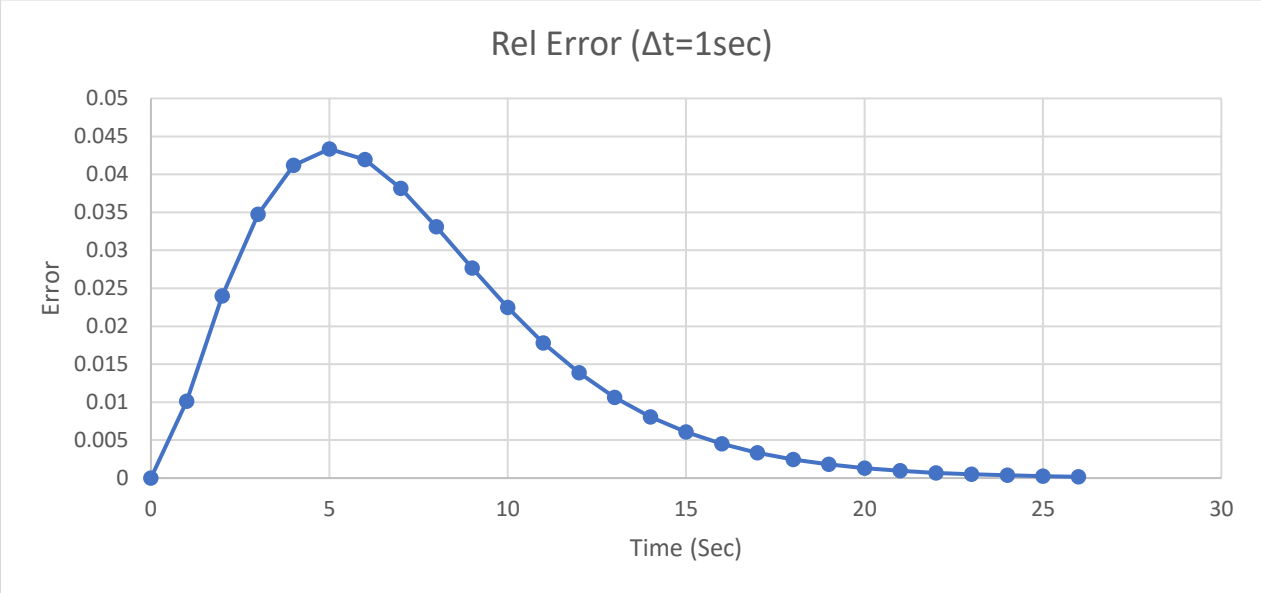
For timestep or delta t = 2secs



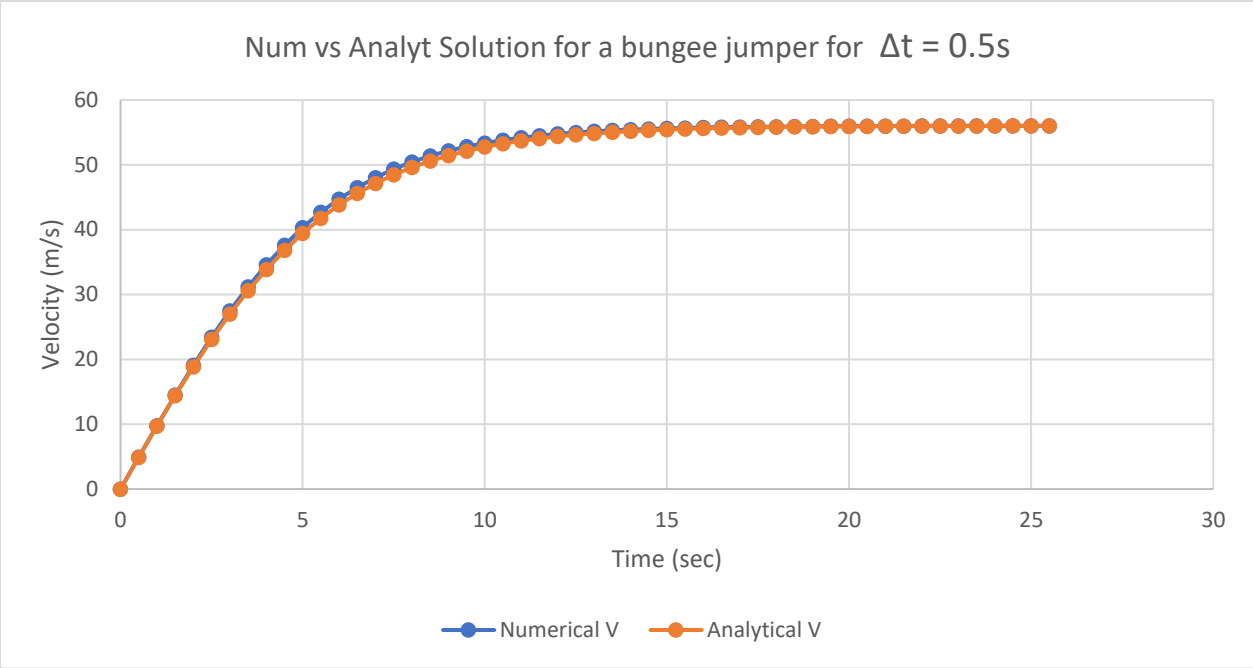


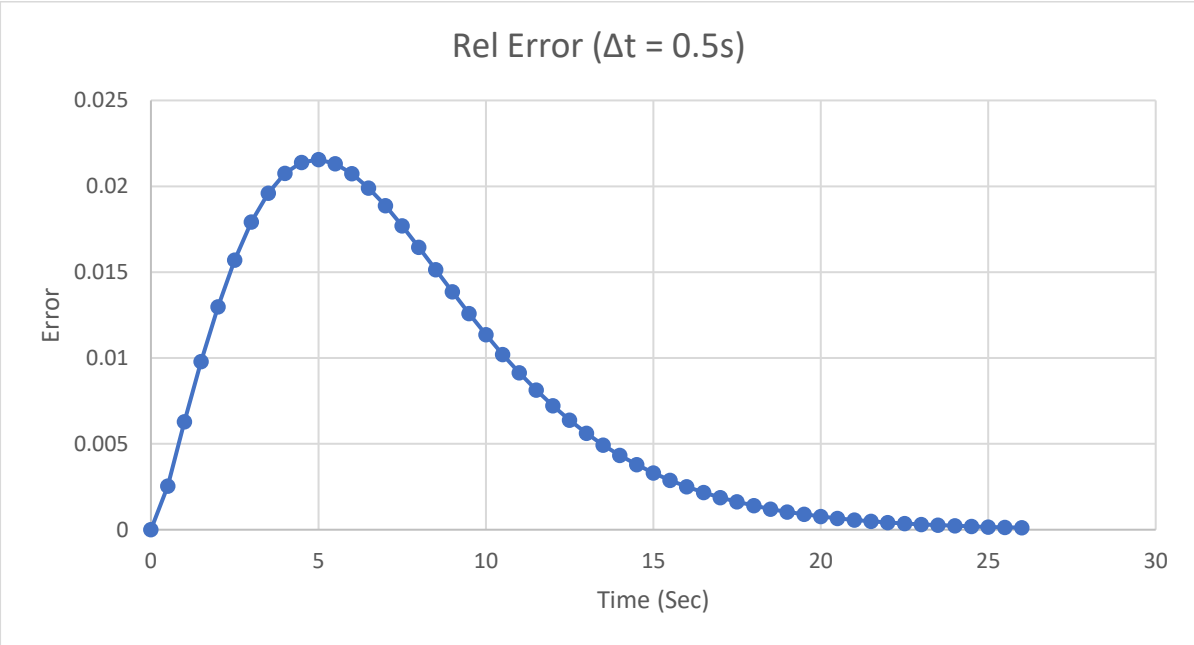
For timestep or Delta t = 1sec:





For timestep or Delta t = 0.5sec:





Sample numerical Problem-

Newton's law of cooling says that the temperature of a body changes at a rate proportional to the difference between its temperature and that of the surrounding medium (the ambient temperature),

$$\frac{dT}{dt} = -k(T - T_a)$$

where T = the temperature of the body (°C), t = time (min), k = the proportionality constant (per minute), and Ta = the ambient temperature (°C).

Suppose that a cup of coffee originally has a temperature of 70 °C. Use Euler's method to compute the temperature from t = 0 to 20 min using a step size of 2 min if T a = 20 °C and k = 0.19/min. Also validate the numerical results/plot with that of analytical solution.

Experiment: Newton's Law of Cooling using Euler's Method

Aim:

To compute the transient temperature of a cup of coffee using Euler's method and to validate the numerical solution by comparison with the analytical solution.

Calculations:

Q. Cooling of a cup of coffee:

→ Given, $T_p = 70°C$

$T_a = 20°C$

$k = 0.19 / min$

* Solving Analytically:

Governing equation :- $\frac{dT}{dt} = -k(T - T_a)$

$\therefore \frac{dT}{(T - T_a)} = -k\,dt$

$\Rightarrow \ln(T - T_a) = -kt + c$ (integrating)

$\therefore T - T_a = e^{-kt} \cdot e^{c}$

$\therefore T - T_a = e^{-kt} \cdot C_1$

Putting in the I.C and B.C

at $t = 0$, $T = T_p = 70°C$

$\therefore 70 - 20 = C_1$

$\Rightarrow C_1 = 50°C$ //

Hence,

$T - 20 = 50 e^{-(0.19) \cdot T}$

* Solving Numerically:

Governing equation :- $\frac{dT}{dt} = -k(T - T_a)$

Approximating $\frac{dT}{dt}$ as $\frac{\Delta T}{\Delta t}$

And by using Euler's method

$$\frac{\Delta T}{\Delta t} = \frac{T_{n+1} - T_n}{\Delta t}$$

Hence,

$$\frac{T_{n+1} - T_n}{\Delta t} = -k\,(T_n - T_a)$$

$$\therefore\ T_{n+1} - T_n = -(k)(\Delta T)\,(T_n - T_a)$$

$$\therefore\ T_{n+1} = T_n - (k)(\Delta T)\,(T_n - T_a)$$

Given Parameters:

Table 1: Physical and numerical parameters used for the transient heat transfer analysis.

Delta t	2
Ta	20
k	0.19
C	50

Calculation Table:

Time	Numerical T	Analytical T	Rel Error
0	70	70	0
2	51	54.19307	0.062609
4	39.22	43.383321	0.106153
6	31.9164	35.990951	0.127663
8	27.388168	30.935594	0.129524
10	24.5806642	27.478431	0.117888
12	22.8400118	25.11421	0.099571
14	21.7608073	23.497411	0.079804
16	21.0917005	22.391744	0.061638
18	20.6768543	21.635622	0.046369
20	20.4196497	21.118539	0.034226
22	20.2601828	20.764925	0.024913
24	20.1613133	20.523103	0.017945

26	20.1000143	20.35773	0.012822
28	20.0620088	20.244638	0.009103
30	20.0384455	20.167298	0.00643
32	20.0238362	20.114409	0.004523
34	20.0147784	20.07824	0.003171
36	20.0091626	20.053505	0.002216
38	20.0056808	20.03659	0.001545
40	20.0035221	20.025023	0.001075
42	20.0021837	20.017112	0.000746
44	20.0013539	20.011702	0.000517
46	20.0008394	20.008003	0.000358
48	20.0005204	20.005473	0.000248
50	20.0003227	20.003743	0.000171
52	20.0002001	20.002559	0.000118
54	20.000124	20.00175	8.13E-05
56	20.0000769	20.001197	5.6E-05

Table 2:Comparison of numerical and analytical temperature values with relative error for Δt = 2 mins

Graphical Analysis:

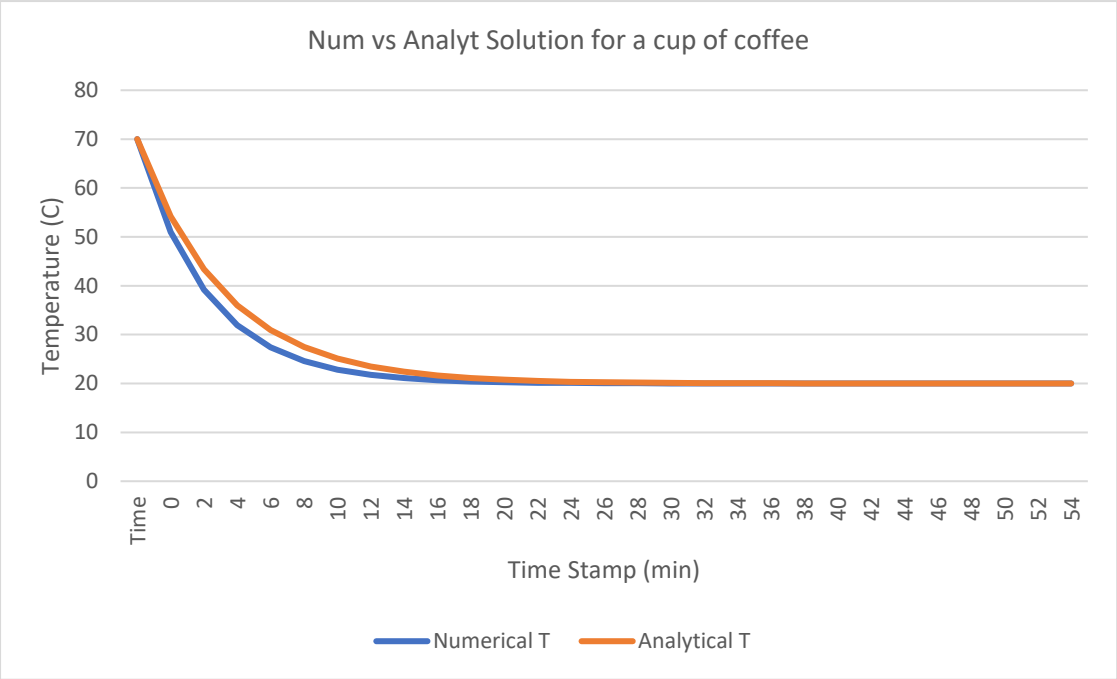


Figure 1:Comparison of Numerical and Analytical Temperature for a Cup of Coffee

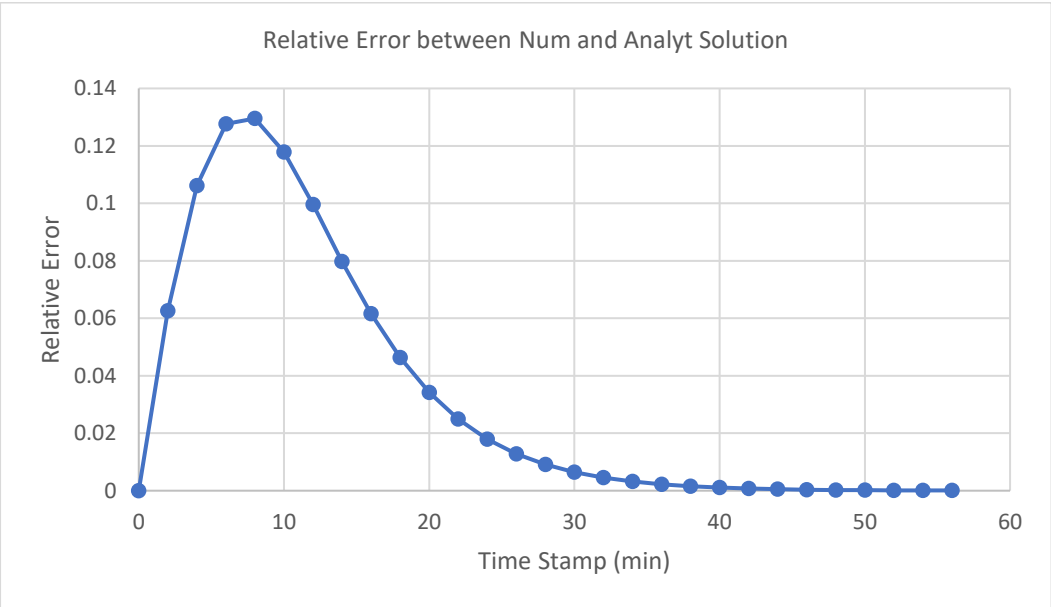


Figure 2: Relative Error between Numerical and Analytical Temperature Solutions